Huawei Access Node KR-ACC-12 Optical Transport Failure Troubleshooting Procedure

Node: KR-ACC-12 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_4548

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve Optical Transport Failure issues on Huawei Access Node KR-ACC-12. The failure domain encompasses SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. The Huawei eNSP (Enhanced Network Service Platform) is a highly scalable, reliable, and secure access platform that supports a wide range of services, including Ethernet, IP, and OTN. The eNSP's Optical Transport Network (OTN) domain is responsible for transmitting optical signals between nodes, and any failure in this domain can result in service disruptions. The Huawei Access Node KR-ACC-12 is a critical component of the OTN domain, and its failure can have a significant impact on the overall network. The OTN domain operates at Layer 1 of the OSI model, using protocols such as ODU (Optical Data Unit) and OPU (Optical Payload Unit) to transmit data. The eNSP's OTN domain is also equipped with advanced features such as OTN switching and grooming, which enable efficient and flexible transmission of optical signals. Therefore, it is essential to troubleshoot and resolve Optical Transport Failure issues on Huawei Access Node KR-ACC-12 in a timely and accurate manner to minimize service disruptions and ensure network reliability.

Pre-requisite Checks

- Verify that the site location is KR Puram Access and the node ID is KR-ACC-12.

- Check the alarm code and severity level to confirm that the issue is related to Optical Transport Failure (ALARM_4548 - Warning).

- Ensure that the Huawei eNSP is running the latest software version.

- Verify that the redundant link is functioning properly and can take over in case of a failure.

- Dispatch remote hands to KR Puram Access to perform physical checks and gather additional information if necessary.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display otn info, Expected Output: OTN information for the node KR-ACC-12, Description: This command displays detailed information about the OTN domain, including the status of optical signals, fiber connections, and SFP/XFP conditions. The output will help isolate the root cause of the failure.

Command: display otn alarm, Expected Output: Alarm information for the node KR-ACC-12, Description: This command displays a list of alarms triggered by the OTN domain, including Optical Transport Failure alarms. The output will help identify the specific issue and its impact on the network.

Command: display otn interface, Expected Output: Interface information for the node KR-ACC-12, Description: This command displays detailed information about the OTN interfaces, including their status, MTU, and CRC error rates. The output will help identify any issues with the interface configuration or hardware.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace otn interface x:y with new-sfp, Description: Replace the SFP/XFP module with a new one to resolve SFP/XFP degradation issues. Verify the new module is properly seated and configured before proceeding.

Step 2: Step: Repair or replace the fiber cable, Command: replace otn interface x:y with new-fiber, Description: Repair or replace the fiber cable to resolve fiber cuts or Rx/Tx power loss issues. Verify the new fiber cable is properly connected and configured before proceeding.

Step 3: Step: Adjust the Rx/Tx power levels, Command: adjust otn interface x:y rx-tx-power, Description: Adjust the Rx/Tx power levels to resolve Rx/Tx power loss issues. Verify the new power levels are within the recommended range before proceeding.

Post-Check Verification

```
Command: display otn info, Expected Output: OTN information for the node KR-ACC-12, Description: Re-run the 'display otn info' command to verify that the issue has been resolved. The output should indicate that the OTN domain is functioning properly and that the SFP/XFP module is operating within normal parameters.

Command: display otn alarm, Expected Output: No alarms triggered by the OTN domain, Description: Re-run the 'display otn alarm' command to verify that the Optical Transport Failure alarm has been cleared. The output should indicate that no alarms are triggered by the OTN domain.

Command: display otn interface, Expected Output: Interface information for the node KR-ACC-12, Description: Re-run the 'display otn interface' command to verify that the OTN interface is functioning properly and that the CRC error rate is within normal parameters.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore otn interface x:y with original-sfp, Description: Restore the original SFP/XFP module if the replacement fails to resolve the issue.

- Step: Restore the original fiber cable, Command: restore otn interface x:y with original-fiber, Description: Restore the original fiber cable if the replacement fails to resolve the issue.

- Step: Reset the Rx/Tx power levels, Command: reset otn interface x:y rx-tx-power, Description: Reset the Rx/Tx power levels to their original values if the adjustment fails to resolve the issue.